

Lepton Mass Ratios from UBT Constants

$$m_\tau/m_\mu = [N_c(\pi-3)^2]^{-1}(1 + \frac{\pi}{2}\alpha) \text{ at } -0.0005\%; \quad m_\mu/m_e = [N_{\text{eff}}(\pi-3)^4]^{-1} \text{ at } +0.27\%$$

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Abstract

Three charged lepton mass ratios are expressed using only UBT-derived constants π , α `[[L1]]`, $N_c = 3$ `[[L0]]`, $N_{\text{eff}} = 12$ `[[L1]]`:

$$\frac{m_\tau}{m_\mu} = \frac{1}{N_c(\pi-3)^2}(1 + \frac{\pi}{2}\alpha) = 16.81695\dots \quad (\text{exp. } 16.81703, -0.0005\%),$$

$$\frac{m_\mu}{m_e} = \frac{1}{N_{\text{eff}}(\pi-3)^4} = 207.327\dots \quad (\text{exp. } 206.768, +0.27\%),$$

$$\frac{m_\tau}{m_e} = \frac{1}{N_c N_{\text{eff}}(\pi-3)^6}(1 + \frac{\pi}{2}\alpha) = 3486.6\dots \quad (\text{exp. } 3477.2, +0.27\%).$$

No free parameter. Status: `[[MC+]]` — structural evidence pending derivation from $S[\Theta]$.

1 UBT Constants Used

$$N_c = 3 \quad \text{[[L0], colour rank]}, \tag{1}$$

$$N_{\text{eff}} = 12 \quad \text{[[L1], su2_twist_neff12.tex]}, \tag{2}$$

$$\alpha^{-1} = 137.035999177549\dots \quad \text{[[L1], alpha_l1_proof.tex]}, \tag{3}$$

$$(\pi-3) = 0.14159265\dots = 2 \cdot 12\pi \Omega_\eta \quad \text{[[L0]]}. \tag{4}$$

The identity $12\pi \Omega_\eta = (\pi-3)/2$ is `[[L0], layer2_kernel_derivation.tex]`.

2 Three Lepton Mass Ratios

Generation structure. Each lepton generation transition adds a factor $(\pi-3)^{-2}$ with a group-theoretic prefactor:

Ratio	Formula	UBT value	Experiment	Error
m_τ/m_μ	$\frac{1}{N_c(\pi-3)^2}(1+\frac{\pi}{2}\alpha)$	16.81695	16.81703	-0.0005%
m_μ/m_e	$\frac{1}{N_{\text{eff}}(\pi-3)^4}$	207.327	206.768	+0.270%
m_τ/m_e	$\frac{1}{N_c N_{\text{eff}}(\pi-3)^6}(1+\frac{\pi}{2}\alpha)$	3486.6	3477.2	+0.270%

Remark 2.1 (Physical interpretation). The factor $(\pi-3)^2$ per generation step arises from the eta-mode occupation: $\pi-3 = 2 \cdot 12\pi \Omega_\eta$ [[**L0**]]. The colour prefactor $N_c = 3$ appears for the τ/μ ratio (the heaviest lepton couples most strongly to the colour background). The mode-count $N_{\text{eff}} = 12$ suppresses the μ/e ratio. The QED one-loop correction $(1+\frac{\pi}{2}\alpha)$ brings the τ/μ ratio from -1.1% to -0.0005%.

What remains open. (1) EW hierarchy $v = 246 \text{ GeV} \ll M_{\text{UBT}}$: blocking the derivation of absolute masses $m_f = Y_f v/\sqrt{2}$. (2) Derivation of the generation structure from $S[\Theta]$: why N_c for τ/μ and N_{eff} for μ/e ? (3) The +0.27% residual in m_μ/m_e and m_τ/m_e : likely a higher-order $(\pi-3)$ or Ω_η correction.

References

- [1] D. Jaros, `alpha_l1_proof.tex`, 2026.
- [2] D. Jaros, `su2_twist_neff12.tex`, 2026.
- [3] D. Jaros, `layer2_kernel_derivation.tex`, 2026.